Natural Language Processing (CO3086)
NLP 252 - CC01: ASSIGNMENT

HO CHI MINH UNIVERSITY OF TECHNOLOGY

Vietnam National University Ho Chi Minh

1 Team Selection

Students are required to form groups of 4 to 5 members and register their groups using the provided group registration system. Each student is allowed to join only one group.

The deadline for group registration is **March 23, 2026**.

2 Task Description

Text preprocessing and tokenization are fundamental steps in Natural Language Processing (NLP), significantly affecting the performance, efficiency, and scalability of language models. Before the emergence of large-scale neural language models, traditional word-level representations dominated NLP pipelines. However, word-level tokenization suffers from several limitations, including large vocabulary sizes and the inability to handle out-of-vocabulary (OOV) words.

To address these challenges, alternative tokenization strategies have been proposed, such as character-level modeling and subword-based methods. Character-level approaches eliminate OOV issues and reduce vocabulary size but often lead to longer input sequences and increased computational cost. Subword tokenization methods, particularly Byte Pair Encoding (BPE), strike a balance between word-level and character-level representations and have become a standard choice for modern language models.

In large-scale language modeling, tokenization plays an even more critical role due to the massive size of training corpora. Well-known benchmark datasets such as One Billion Word Benchmark, WikiText-103, Text8, and Enwik8 are commonly used to evaluate the effectiveness of different preprocessing and tokenization techniques. Each dataset presents unique challenges in terms of vocabulary size, sequence length, and linguistic diversity.

This assignment focuses on analyzing and comparing different text preprocessing and tokenization methods, including word-level, character-level, and BPE-based approaches, applied to large-scale text datasets. Students will investigate how these methods affect vocabulary size, sequence length, training efficiency, and downstream language modeling performance.

3 Submission Instructions

Students are required to write a report and conduct experiments on text preprocessing and tokenization techniques. The assignment consists of both theoretical analysis and empirical evaluation.

The report should include:

- An overview of text preprocessing in NLP and its importance in language modeling.

- A detailed explanation of word-level, character-level, and BPE-based tokenization methods.
- A comparative analysis of tokenization methods in terms of vocabulary size, sequence length, and computational efficiency.
- Experimental results obtained by implementing and evaluating the proposed methods on all provided datasets, including One Billion Word, WikiText-103, Text8, and Enwik8.
- An evaluation of the impact of different tokenization strategies on language modeling performance (e.g., perplexity, training speed).
- Insights, limitations, and conclusions drawn from the experiments.

Students must implement the preprocessing pipelines and tokenization methods themselves or use well-established libraries, clearly describing their implementation details. Final submissions should include:

- A written report in PDF format.
- Source code for data preprocessing, tokenization, and experiments.

The deadline for assignment submission is **April 15, 2026**.